

Enes Senel

Senior Scientist, Computational Biology — single-cell multi-omics · machine learning · spatial omics

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PROFILE

Computer scientist turned computational biologist. I lead single-cell multi-omics programs in oncology and immunology at Johnson & Johnson Innovative Medicine (Berlin), and I've spent ten years applying machine learning to hard problems across biology and medicine — from robotic biopsy guidance to spatial transcriptomics to clinical-stage cell therapy. I build methods and pipelines that other people actually use: Optocoder and novoSpaRc are cited and run by labs worldwide, and my J&J pipelines are the team's default infrastructure.

CORE EXPERTISE

Single-cell & spatial omics: scRNA-seq, CITE-seq, spatial transcriptomics, multi-modal integration, biomarker discovery.

Machine learning: graph neural networks, deep learning (PyTorch/TensorFlow), optimal transport, protein language models, computer vision.

Engineering: production Python pipelines, Nextflow, containerisation, reproducibility standards, C++/CUDA.

Leadership: analytical strategy across parallel programs, mentoring, managing consultants, cross-functional work with clinical, discovery, and CMC teams.

EXPERIENCE

Senior Scientist, Computational Biology — Johnson & Johnson Innovative Medicine, Berlin

Aug 2025 — Present

- Technical lead for single-cell multi-omics in oncology and immunology, from discovery through product development.
- Lead programs feeding biomarker discovery, mechanism-of-action work, and translational strategy for pipeline assets.
- Set analytical direction and standards across parallel projects; mentor junior scientists and manage external consultants.

Postdoctoral Data Scientist — Johnson & Johnson Innovative Medicine, Berlin

Oct 2023 — Jul 2025

- Built the team's production single-cell infrastructure: automated Python pipelines for scRNA-seq and CITE-seq, replacing manual workflows.
- Led single-cell analyses for biomarker discovery in a clinical-stage cell therapy and for CMC process characterisation.

Doctoral Researcher — Max Delbrück Center for Molecular Medicine, Berlin

Oct 2019 — Sep 2023

- Created Optocoder (ML decoding of imaging-based spatial transcriptomics; NAR Genomics & Bioinformatics) and co-led novoSpaRc (Nature Protocols) — both widely used open-source tools.
- Applied graph neural networks to spatial omics data, modelling cells as graphs over tissue structure.
- Co-inventor on a US patent for 3D spatial gene-expression reconstruction.

Earlier roles — Coriolis Pharma · Max Planck Institute · Ozyegin Robotics Lab

2014 — 2019

- Deep learning for pharmaceutical QC (Coriolis); multi-omics integration in neuroscience (Max Planck); real-time ultrasound needle tracking for a biopsy robot (Ozyegin, five papers).

EDUCATION

PhD Computational Biology — Max Delbrück Center / Humboldt University, Berlin (2019–2026, expected)

MSc Computer Science — Technical University of Munich (2019)

BSc Computer Science — Ozyegin University, Istanbul (2016)

SELECTED PUBLICATIONS & OUTPUTS

- Benchmarking single-cell foundation models for real-world RNA-seq data integration — bioRxiv (2026).
- Benchmarking Next-Generation Sequencing Platforms: Ultima UG 100 vs. Illumina NovaSeq X Plus — bioRxiv (2026).
- Optocoder — NAR Genomics & Bioinformatics (2022); novoSpaRc — Nature Protocols (2021, co-first author); Gene Expression Cartography — Nature (2019).
- US patent (co-inventor): 3D spatial gene-expression reconstruction. Five robotics papers (ICRA ×2, ICAR, Mechatronics, TJEECS).
- Open source: github.com/rajewsky-lab/optocoder · github.com/rajewsky-lab/novospaarc.

Full publication list, project stories, and contact details: enessenel.com